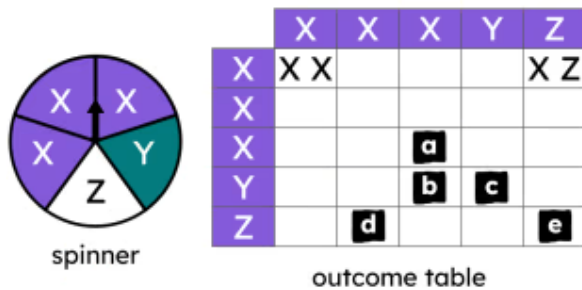




## Problem solving with possible outcomes

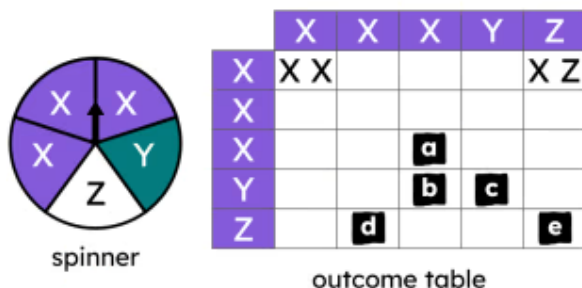
- 1 Trial: The spinner is spun twice. The order of the two outcomes doesn't matter. The incomplete outcome table shows all outcomes of the trial. Match each letter the sample space to its value. Write the correct letter in each box



|   |   |
|---|---|
| a | a |
| b | b |
| c | c |
| d | d |
| e | e |

|   |     |
|---|-----|
| e | Z Z |
| a | X X |
| c | Y Y |
| d | Z X |
| b | Y X |

- 2 Trial: The spinner is spun twice. The order of the two outcomes doesn't matter. Which of these statements are correct for this trial? Tick 2 correct answers



- ☒ One outcome is more likely to happen than any other outcome.
- ☐ Two equally likely outcomes are both the most likely to happen.
- ☐ Two equally likely outcomes are both the least likely to happen.
- ☐ The most likely outcome is three times as likely as the least likely outcome.
- ☒ The most likely outcome is nine times as likely as the least likely outcome.



- 3 Trial: The product of two spins of the spinner. This outcome table represents all outcomes of the trial. Match the start of each statement to the correct end of the statement. Write the correct letter in each box

| ×  | 0 | 3  | 3  | 3  | 8  | 10  |
|----|---|----|----|----|----|-----|
| 0  | 0 | 0  | 0  | 0  | 0  | 0   |
| 3  | 0 | 9  | 9  | 9  | 24 | 30  |
| 3  | 0 | 9  | 9  | 9  | 24 | 30  |
| 3  | 0 | 9  | 9  | 9  | 24 | 30  |
| 8  | 0 | 24 | 24 | 24 | 64 | 80  |
| 10 | 0 | 30 | 30 | 30 | 80 | 100 |

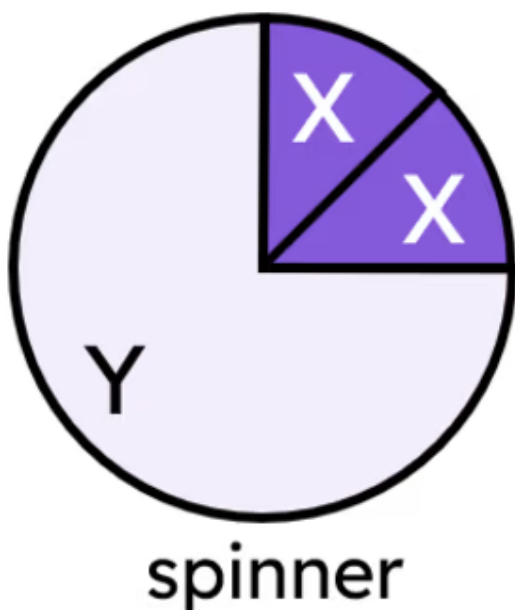
outcome table

Event:

- Alex wins if the outcome is greater than 9.
- Sam wins if the outcome is less than 9.
- Alex and Sam draw if the outcome is 9.

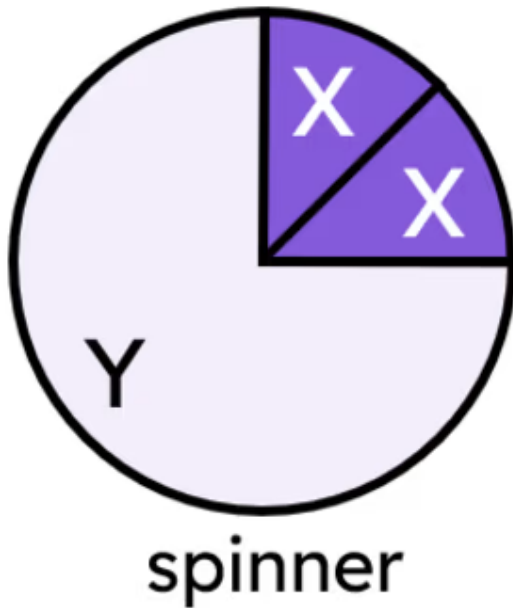
|   |                             |   |                                                                        |
|---|-----------------------------|---|------------------------------------------------------------------------|
| a | The event that Alex wins... | c | is the least likely outcome.                                           |
| b | The event that Sam wins...  | a | is the most likely outcome.                                            |
| c | The event that they draw... | b | is in between the least and most likely outcomes on a likelihood scale |

- 4 Trial: The spinner will be spun once. Which of these statements about one stage of this trial is correct? Tick 2 correct answers



- ☐ The spinner is twice as likely to land on X compared to Y.
- ☐ The spinner is twice as likely to land on Y compared to X.
- ☒ The spinner is three times as likely to land on Y compared to X.
- ☐ It is possible to split this spinner into 6 equally likely outcomes.
- ☒ It is possible to split this spinner into 8 equally likely outcomes.

- 5 Trial: The spinner will be spun twice, and the order of the two stages does not matter. Match the start of each statement to the correct outcome. Write the correct letter in each box



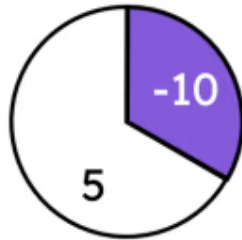
|   |                                 |
|---|---------------------------------|
| a | The most common outcome is...   |
| b | The least common outcome is...  |
| c | Y Y appears 1.5× as often as... |

|   |     |
|---|-----|
| a | Y Y |
| c | XY  |
| b | X X |

- 6 Trial: Spinner A and B are spun once each. The sum of the two outcomes is found. Which of these statements about the outcomes of this trial are correct? Tick **3** correct answers



spinner A



spinner B

- ☒ 12 is the most likely outcome.
- ☐ -13 is the most likely outcome.
- ☐ The most likely outcome is 4 times as likely than the least likely outcome.
- ☒ The most likely outcome is 8 times as likely than the least likely outcome.
- ☒ The most likely outcome is twice as likely than the second most likely outcome.

